

TM Forum Introductory Guide

Autonomous Network Levels Evaluation Methodology

IG1252

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1. Terminology

1.1. Terms

Term	Description
Effectiveness indicator	An effectiveness indicator is a metric that measures the actual effectiveness of automation and intelligence capabilities introduced to network operation.
Operation flow	An operation flow is an encapsulation of one or more management tasks.
Task	The embodiment of the key autonomy capability in the network, is the smallest unit divided in the operation flow domain.
Technology maturity model	Technology maturity model aims at the step-by-step construction of system cognitive ability and defines the level requirements and corresponding interface requirements for operation systems from the perspective of technology.
Note: The terms defined in IG1230 are applicable to this document.	

2. Scope of Document

This document defines the Autonomous Networks Level based on TM Forum IG1218, IG1230, and serves as the autonomous level evaluation component of the Autonomous Operations Maturity Model. It describes the concepts of Autonomous Networks Level, which include Autonomous Networks Level methodology and approach that consists of technology maturity model and key effectiveness indicators, operational processes, their underlying sub-processes and tasks, task evaluation criteria, scoring method etc., and finally establishes a standardized evaluation approach for assessing (in an offline manner) the Autonomous Networks Level of a network, or part of a network.

3. Background, Benefits, and Usage

3.1. Background and benefits

A common understanding of the levels of autonomous networks will be helpful to the telecommunications industry. The operator's operational and management efforts and dedicated human resources will vary according to different autonomous networks levels. It would be beneficial for operators to have clear view on the expectation of the level of their network (s) to measure the maturity of the network autonomy, so that they could firstly focus on the important features which should be prioritized to achieve a certain level of autonomy in their networks. This approach will help operators to smoothly migrate to a higher autonomy level and a better operational efficiency.

Several standards development organizations (SDOs) have been trying to assess the different levels of network automation across different service providers and enterprise businesses. However, the reference guidelines in systematic evolution of automation and quantized measurement methods are still unclear.

It is essential to develop a clear, explicit, and operable Autonomous Networks Level methodology and approach, which is the purpose of this document. It has the following benefits:

- Providing a basis for measuring the level of an autonomous network, or autonomous network feature, along with its components and workflows, with detailed process-oriented approach and effect-oriented metrics to achieve quantized evaluation.
- Providing reference for gap identification and priority analysis for operators, vendors and other participants of the telecommunications industry for roadmap planning of future work on improving network autonomy.

3.2. Usage of this document

The document is intended to be used by service providers help them in conducting high level autonomy assessments of their existing network, service and system, identifying their maturity and specific areas for future improvement, as well as by solution providers as a reference for the development direction of emerging network innovations.

For example, the operator can determine the level criteria of L0-L5 of fault identification based on the level characteristics and perform level autonomy assessment.

The baseline process structure used within the methodology may vary. A service provider can, if required, adapt the flows or structures to suit their own environment. The evaluation methodology remains fully valid, thus has broad applicability.

4. 1. Autonomous Network Level Methodology

4.1. Background

An autonomous network level framework, inspired by '[1]' and '[2]' is defined in IG1230³, which provides the background for exploring the path of autonomous network level evaluation and evolution.

Table 4-1 Autonomous network level framework

Autonomous Levels	L0: Manual Operation & Maintenance	L1: Assisted Operation & Maintenance	L2: Partial Autonomous Networks	L3: Conditional Autonomous Networks	L4: High Autonomous Networks	L5: Full Autonomous Networks
Execution	P	P/S	S	S	S	S
Awareness	P	P/S	P/S	S	S	S
Analysis	P	P	P/S	P/S	S	S
Decision	P	P	P	P/S	S	S
Intent/Experience	P	P	P	P	P/S	S
Applicability	N/A	Select scenarios				All scenarios

P : People (manual) ; S : Systems (autonomous)

Note 1: System includes management system, O&M tools and network.

Note 2: The table 4-1 is consistent with IG1218 and may be adjusted in the future version.

4.2. Methodology

4.2.1. Principle

The design of AN level evaluation methodology follows three principles:

- The ultimate goal of level evaluation is to promote and guide the construction and deployment of autonomous networks with regard to the objectives of increasing OAM efficiency to reduce costs and improving customer experience to increase business income, level evaluation serves the purpose to measure the effect of investment.
- Autonomous network evolution is not to replace the existing network with a completely new one, instead it targets at enabling capability enhancements in different layers and different domains of a production network. The actual system should be taken and decomposed as a set of evaluation objects to obtain the evaluation result, which can be used as a reference for the guidance of subsequent capability enhancement and level improvement.
- The basic principles of the operations of autonomous domains are autonomy of individual autonomous domains and collaboration of cross-autonomous domains. The former requires assessment of intra-domain closed loop autonomous capabilities, while the latter imposes requirements on cross-domain interfaces.

4.2.2. Methodology

Derived from the above principles, in order to comprehensively and accurately evaluate the autonomous network level, the overall autonomous network level methodology consists of the following three parts:

1. **Effectiveness indicators:** A set of effectiveness indicators are defined to measure the actual deployment effectiveness of AN, which describe the actual performance and effect of autonomous capability after it is introduced into the system, helping service providers to further understand the benefits of the system autonomy upgrade from the perspectives of operation, customers and business.
2. **Technology maturity model:** The technology maturity assessment on specific capabilities in the network is established to ensure the accuracy and operability of evaluation results by refining the evaluation and measure of autonomy. Each evaluation object shall correspond to the specific responsible subject in practice and measures can be taken against weakness of system.
3. **Resource closed loop autonomy:** Evaluate the autonomous capability in a resource closed loop to guide the level-by-level upgrade of the domain specific self-x capabilities.

4.2.2.1. Effectiveness Indicators

The level evaluation describes the autonomous level of the established business process or autonomous domain, but it is not sufficient to reflect the actual deployment effect of AN.

To promote customer experience and OAM efficiency, based on the zero-X and self-X vision, an effectiveness indicator system should cover the whole life cycle of network and could be easily and directly obtained from the system. Principles of the selection of Effectiveness Indicators are as follows:

1. **Positive correlation:** Effectiveness indicators positively guide the improvement of autonomous network capabilities, that is, the increase of effectiveness indicators is positively correlated to the improvement of autonomous network.
2. **Objectivity:** The value of effectiveness indicators is not influenced by human subjective factors, or the subjective impact is controllable or identifiable.
3. **Quantifiable:** Effectiveness indicators should be measurable and comparable

It is described in IG1256 for more details.

4.2.2.2. Technology maturity model

During the closed-loop execution inside any autonomous domain, the achievable autonomous level of system autonomy (i.e. the separation of duty interface between people and system) is ultimately dependent on the system's cognitive ability to comprehend and react based on the underlying business or service or domain specific logic of closed-loop execution towards external environment and realize adaptation to changes.

According to the Data-Information-Knowledge-Wisdom or DIKW^{4,5} system, (see appendix), a knowledge management model (pyramid hierarchical system of data, information, knowledge and wisdom), the cognitive ability of the system is gradually improved: from "know nothing", "know what", "know how", "know why" to "know all", and finally complete autonomy is realized.

Hence, referring to the DIKW system and aiming at the step-by-step construction of system cognitive ability, the technical maturity model are defined: from offline manual implementation, online recording, statically configured rules driven, dynamically programmable policies driven, AI-assisted evolution, to full autonomy.

4.2.2.3. Resource closed loop autonomy:

The evaluation methodology of resource closed loop autonomy is from the perspective of human-system work division: depending on those tasks can be performed by the telecom system, those tasks to be performed by human, and those tasks that are being performed by cooperation of human and telecom system. Starting from a fully manual operation, the autonomy of a resource closed loop gradually increases as more tasks are performed by telecom system instead of human and reaches fully autonomy when all tasks are performed by the system.

4.2.3. Characteristics of AN level

The AN level evaluation approach adopts the characteristics of levels applicable to all operation layers. It takes the theory of human-system cooperation as a reference and uses the technology maturity model as a guide to define the characteristics of L0-L5.

In order to facilitate practice, the specific characteristics of each level are defined for IT systems and resource closed loop respectively, which can be applied in an independent or combined manner, according to the specific objective of the user in context.

4.2.3.1. General technical capability level characteristics

Table 4-2 General technical capability level characteristics

Level	Typical characteristics	Interface requirements	Interface requirements description
Level 0- Manual management	Offline manual implementation	Manual operation interface	The interface shall be called manually by the upper layer.
Level 1- Assisted management	Manual implementation, online recording	Event reporting interface	The system shall respond to requirements for online recording and record the event reporting.
Level 2-Partial autonomous network	Automation driven by statically configured rules	Rule configuration interface	The system shall respond to requirements for static rules and support corresponding rules configuration capabilities.
Level 3- Conditional autonomous network	Automation implementation driven by dynamically programmable policies	Policy design and configuration	The system shall respond to requirements for dynamically programmable rules (policies) and support corresponding design and configuration capabilities.
Level 4-High autonomous network	Automatic implementation driven by AI assisted (if needed) knowledge, with continuous	AI Model management	The system shall respond to requirements for rules/policies generation, and provide corresponding local adaptation, model training and inference capabilities, with continuous learning and rapid evolution.

Level	Typical characteristics	Interface requirements	Interface requirements description
	learning and rapid evolution.		
Level 5-Full autonomous network	Automatic implementation capable of self-evolution and adapting to changes.	Knowledge Closed loop management	Corresponding to autonomous evolution, knowledge (rules, policies, etc.) and AI/ML models are automatically generated based on AI deep learning, without human interference. The system shall respond to the closed-loop management requirements for knowledge and AI models.

4.2.3.2. Resource closed loop level characteristics

To evaluate the autonomy of resource closed loops, first, it is required to confirm whether the general technical capability level characteristics requirements and the supportive interface requirements for the high-level closed loop are met according to Table 4-2 and Table 4-3 respectively. On the other hand, aggregated autonomy of a resource closed loop can be evaluated in a black-box manner according to Table 4-4.

Table 4-3 Interface requirements of general technical capability level characteristics for resource closed loops

Level	Interface requirements	Interface requirements description
Level 0-Manual management	Manual operation interface	The interface shall be called manually by the upper layer.
Level 1-Assisted management	Event reporting interface	The resource closed loop shall respond to requirements for online recording and on-demand reporting of specified events.
Level 2-Partial autonomous network	Rule configuration interface	The resource closed loop shall respond to requirements for executing statically configurable rules and provide corresponding configuration capabilities.
Level 3-Conditional autonomous network	Policy configuration interface	The resource closed loop shall respond to requirements for interpreting and executing dynamically programmable rules (i.e., policies) and provide corresponding configuration capabilities.
Level 4-High autonomous network	AI Model management interface	The resource closed loop shall respond to requirements for managing AI models used for intelligent dynamic automation adaptation capabilities (e.g., rules generation), and provide corresponding local adaptation, model training and model

Level	Interface requirements	Interface requirements description
		reference capabilities based on the AI models managed through interfaces. In certain scenarios, the resource closed loop may respond to requirements expressed as intent from the service operation layer.
Level 5-Full autonomous network	Intent driven AI/Knowledge closed loop management interface	Knowledge (rules, policies, etc.) and AI/ML models are automatically generated based on AI deep learning, without manual participation. The resource closed loop shall respond to the AI/Knowledge closed-loop management requirements for end-to-end autonomous knowledge devops and MLOps, including automatic generation, application monitoring, and continuous tuning etc. that covers the whole life-cycle of the intelligence/AI governance and provide the intent-driven interfaces to guide the self evolution capability towards the desired expectations.

Table 4-4 Level characteristics of resource close loops

Level	Typical characteristics	Expected Effects
Level 0-Manual management	Manual O&M	Awareness, analysis, decision, execution and intent are fully manually accomplished. No manpower is saved.
Level 1-Assisted management	Machine assisting humans: The system assists humans to perform awareness and execution tasks.	Awareness and execution are assisted by system. Actions of awareness, execution and intent/experience can be online recorded and traced. Part of manpower can be saved, and more manpower can be focused on analysis and decision.
Level 2-Partial autonomous network	Machine takes Repetitive tasks: The system provides simple and repetitive batch processing capabilities for task execution.	Execution is automated. Some awareness and analysis functions can be accomplished for certain scenarios based on static (human predefined) rules/policies. Part of manpower can be saved from simple and repetitive work on data collection, pre-processing and analysis.
Level 3-Conditional autonomous network	Machine fully on-duty: The system dynamically and automatically detects real-time network status,	Awareness, execution and most analysis are automated. Scenario decoupled common rules/policies/AI models which can be dynamically updated. Some automatic decision-

Level	Typical characteristics	Expected Effects
	changes, and exceptions, and provides automatic decision-making capability in simple scenarios	making can be achieved under human predefined rule/policy.
Level 4-High autonomous network	Machine self-adaptation: Based on the O&M intent, the system has the self-adaptive closed-loop processing capability in specific scenarios.	Awareness, analysis, decision, and execution are automated, intent/experience driven closed-loop control can be accomplished for certain scenarios under human supervision in emergency situations. Most of the manpower can be focused on expert experience, AI model management, and novel or unsupported scenarios.
Level 5-Full autonomous network	Machine self-evolution: The system is capable of self-learning and self-evolving for new scenarios, achieving complete autonomy in all scenarios.	Awareness, analysis, decision, execution and intent/experience are autonomous. The whole autonomy mechanisms including expert experience and AI model management can be auto-evolved for all scenarios. Manpower is only needed to generate intent and monitor the progress and status.

5. Autonomous Network Level Evaluation

5.1. Procedure

5.1.1. Overview

This procedure consists of 3 steps as below and it will be described in following sections:

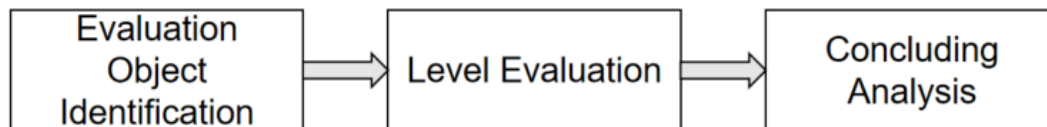


Figure 5-1. Procedure of Autonomous Network Level Evaluation

- Evaluation object identification: identify the scope and content of evolution.
- Level evaluation: specific evaluation method of general technical capability and resource closed loop.
- Concluding analysis: output and analyze the result.

5.1.2. Evaluation Object Identification

The evaluation object is a service and/or network with selected operation flow for autonomous network level evaluation.

Determine the evaluation objects based on selecting an operational use case from three-dimensional elements of services domain, network technology domain, and operational flows shown in Figure 5-2.

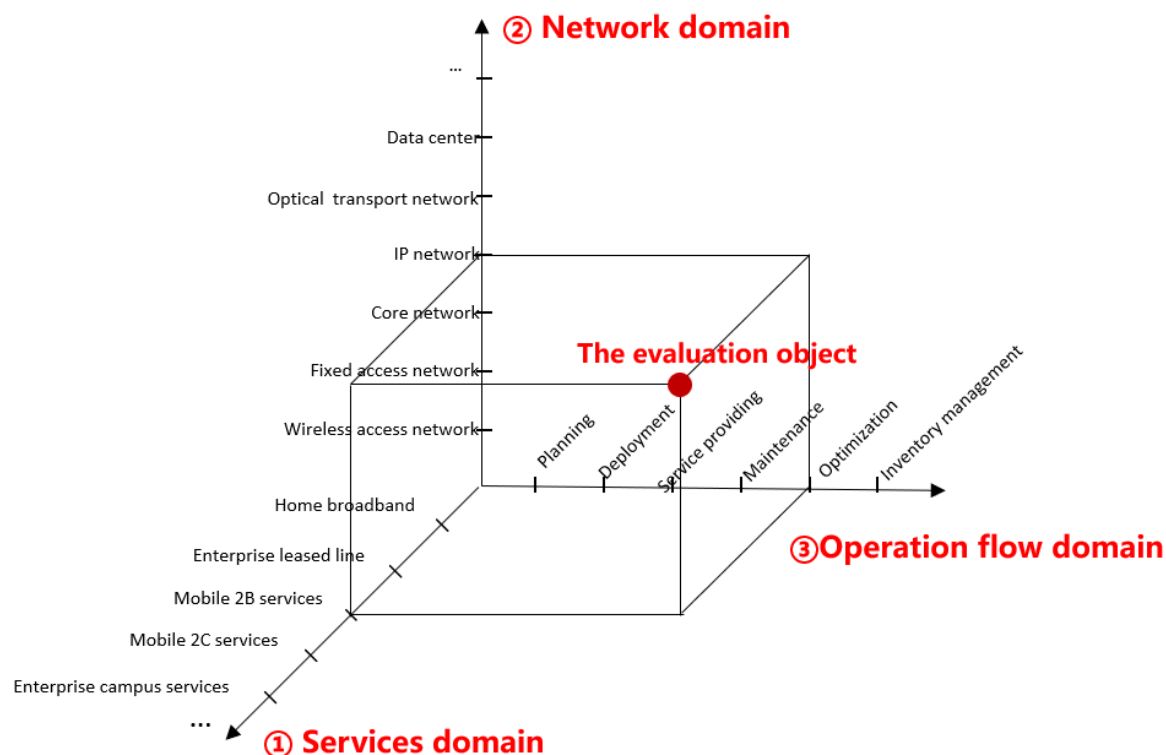


Figure 5-2. Identify the evaluation object

Operation flows shall follow the principle of authenticity, which reflects the flows of the evaluation object in an actual production environment. In business/service/ resource operation layer, the operational flows include: Planning, Deployment, Service providing, Maintenance, Optimization and Inventory Management.

Each production activity can be mapped to a unique, or atomic, operation task. One production activity-to-many operations task cannot exist. If one production activity is mapped to multiple standard operation tasks, the production activity needs to be further decomposed. This decomposition should continue until it is no longer possible to further break down items.

Task is identified according to the operator's existing network management lifecycle and the services involved in the effectiveness indicator system. For a particular layer, each contextualized task corresponds to one and only IT system/management responsible subject.

5.1.3. Level Evaluation

In the actual evaluation process, the evaluation object needs to further focus on the network or service dimension and refine the context of the task from the perspectives of network standards, resource object types, service types and so on. Each task is divided to form specific task contextualization, meaning the evaluation object with specific context, also the smallest scoring unit in the whole level evaluation. Assess the level of the task contextualization based on the AN level characteristics. Valid scores are between 1.0 (manual operation) and 5.0 (full autonomy).

After technical assessment of each task contextualization, the average score of all task contextualization included in each task is calculated as the level of this task. The level of evaluation object is the average score of all tasks included in the specific operation flow. For resource closed loop, specific score aggregation methods exist according to Table 4-4.

5.1.4. Concluding Analysis

Combined with the effectiveness indicators, advantages and weakness of the system are determined according to the assessment result. As each contextualized task clearly corresponds to a management responsibility subject/IT system, which is responsible for the continuous improvement and subsequent enhancement of this task.

5.2. Evaluation Object Identification and Decomposition

5.2.1. Operation flow and sub operation flow

Based on service provider perspectives and best practices, each operational layer of the autonomous network can be decomposed to the following operation flows:

- End-to-end operation flows in business layer: Including customer lifecycle management processes and product/offering lifecycle management processes.
- End-to-end operation flows in services layer: Including the flows of resource-facing services planning, deployment, maintenance, optimization and inventory management . Also, includes the flows of customer-facing services planning, deployment, service providing, maintenance, and optimization.
- Domain specific supportive sub flows in resource layer: Provide support to the operation flows of upper layer, including planning, deployment, maintenance, and optimization.

Planning, deployment, service providing, maintenance, and optimization compose the whole life cycle process of the production. Inventory management is an important foundation for operators in network O&M, and also a general inventory management process may be used and embodied in the other 5 flows. Based on the requirement of operators for the building and evaluation of the operation flows, the logical relation of flows is put forward as Figure 5-3.

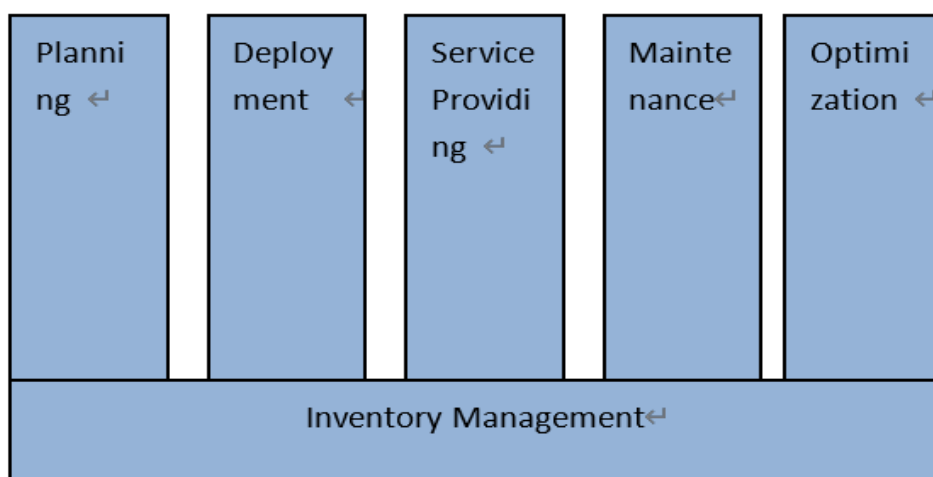


Figure 5-3. Logic relation of flows

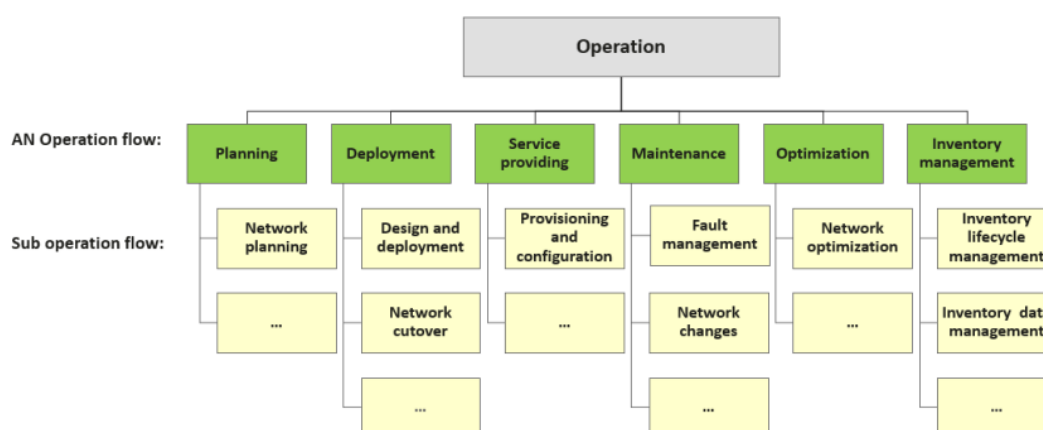


Figure 5-4. AN operation flow

The AN operation flows definition may cover business layer, services layer and resource layer of AN network.

Table 5-1. AN operation flow

AN Operation Flow	AN Operation Flow Description
Planning	Based on the market requirements, forecast and analyze products, services and network requirements, to form service and network planning. Include: 1) Based on the customer's business requirement, service development objective, network construction plan, and network capacity analysis and prediction, output the planning solution.

AN Operation Flow	AN Operation Flow Description
	2) Based on the planning solution, the services and network survey, technical requirements of the solution, output the high level design.
Deployment	Design, install, deliver, and verify the service and network based on the planning to complete the preparations before service provisioning. Including: 1) Based on the planning, complete hardware and software detail design, installation and optimization, as well as services and network initial configuration. 2) Output acceptance reports, pass the acceptance criteria and ready for service and network provision and maintenance
Service Providing	Configure services and networks based on service requirements of customers. Monitor the service quality, quickly respond to customer complaint on service quality issues and recover the service quality to acceptable level in a timely manner
Maintenance	Based on customer's SLA requirements, monitor and manage service quality or network operation status, find and solve related problems or faults (including hidden faults) in time, and ensure that service quality and network operation are at state.
Optimization	To improve service quality and network performance, analyze services KQI and network KPIs and adjust services and network configurations to ensure optimal service and network running. This includes: 1) Analyze network status and service quality trend, performance tests, customer complaints/feedback, and resource utilization. 2) Formulate and implement services and network optimization solutions, to meet customer service experience and resource utilization requirements
Inventory Management	Establish, manage and administer the enterprise's resource inventory, as embodied in the Resource Inventory Database, and monitor and report on the usage and access to the resource inventory, and the quality of the data maintained in it.

Using these best practices, we can then decompose the AN operation flows into sub operation flows.

Note: The sub operation flows mainly focus on services layer and resource layer in document version, further detail will be added in later document versions.

Table 5-2. Sub operation flow

AN Operation Flow	Sub Operation Flow	Description of Sub Operation Flow
Planning	Network planning	Analyze network expansion requirements, calculate network expansion goals, formulate network adjustment plans based on the current network status, and formulate network construction projects.
Deployment	Design and deployment	Design, develop, installation, and verify the service and network resources
	Network cutover	Migrate subscribers from other networks to the new network so that the new network enters the production process and provides services to subscribers, and network status information changes are synchronized to inventory management.
Maintenance	Fault management	Set monitoring rules based on the customer's SLA requirements, monitor the services quality and network status in real time, detect faults or potential risks in a timely manner, demarcate and locate the faults, analyze the root causes, and rectify the faults or potential risks, to ensure the normal running of the network and provide reliable service quality.
	Network change	Network change requests cover many cases, such as equipment replacements, network topology adjustment, software upgrades or patches, etc. These changes are generated as a result of maintenance. Taking into account impact on user services, network change plans are formulated (such as change time, network elements, operation steps, verification plans, fallback plans, etc.), and changes are implemented. Network status information changes are synchronized to inventory management.
	Routine maintenance	In order to ensure the normal operation of the network, routine checks and maintenance are carried out on the operation status of the network. Routine maintenance is mainly completed through the formulation,

AN Operation Flow	Sub Operation Flow	Description of Sub Operation Flow
		implementation, and inspection of regular operation plans.
	Test	Verify the corresponding indicators of service connectivity, service SLA, network KPI, etc. to ensure the normal operation of the network.
Optimization	Network Optimization	Based on the collection and processing of data related to network operation, statistical analysis and data mining, the network operation status is evaluated in multiple dimensions such as network operation and cost efficiency, and hidden network operation risks are discovered. Based on the current network operation status, formulate network optimization plans, implement network optimization plans and evaluate optimization results. Control the entire network optimization process.
Service Providing	Provisioning and configuration	According to service requirements, schedule and arrange service processes and resources, design corresponding network plans. Based on the plan, adjust operation and network configuration
	Customer complaint (related to services)	Specially monitor the service quality, discover the customer's service quality problems in a timely manner, and quickly respond to them. At the same time, the service SLA is monitored and managed, and the risk of SLA violations is timely warned. To locate and analyze the problem of customer complaints, deal with the problems caused by the network and follow up the results, ensure the resolution of customer complaints, and improve customer satisfaction.
Inventory Management	Inventory lifecycle management	Determine, provide and maintain the inventory management of corresponding hardware, software and supporting equipment and facilities, and enable inventory alterations in time according to the user needs such as software updates, topology changes
	Inventory data management	Manage existing network data resources, record the type and quantity of resources, the relationship between resources, topology and other

AN Operation Flow	Sub Operation Flow	Description of Sub Operation Flow
		information, so as to improve the efficiency of process and resource use.
Note: The flows can be expanded according to requirements.		

5.2.2. Operation flow decomposition and task identification

5.2.2.1. Relation between task and flow

A task is the embodiment of the key autonomy capability in the network, is the smallest unit divided in the operation flow domain. The autonomy capabilities of the tasks within the operation flow may impact the network autonomy level.

Note: Categorization of tasks described in V1.1.0 of this document is specific to resource closed loop, thus is moved to Section 5.3.2.2.1.

5.2.2.2. Planning flow decomposition and task identification

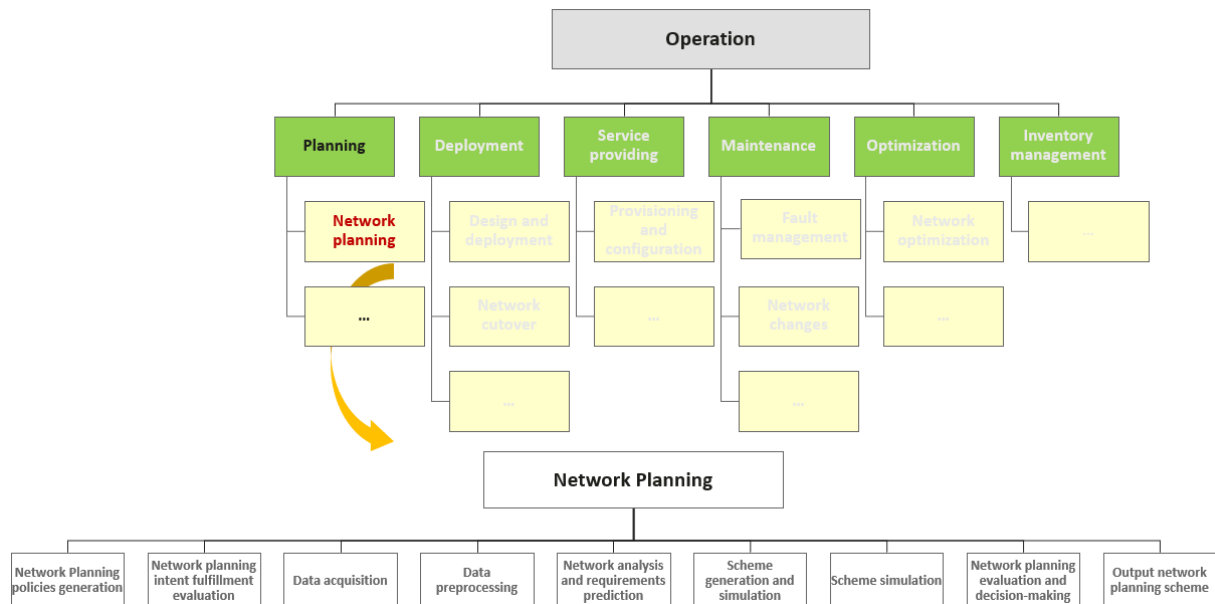


Figure 5-5. Operation tasks of network planning

Table 5-3. Operation tasks of network planning

Sub Operation Flow	Operation Task	Operation Task Description
	Network Planning policies generation	According to customers' business intentions, business/service development goals, and network construction plans, output network construction strategies (such as coverage area priority, deployment frequency band priority, network architecture priority) and network construction standards (such as coverage/rate/delay).
	Network planning intent fulfillment evaluation	Evaluate and confirm the results of the network planning and evaluate the satisfaction of the

Sub Operation Flow	Operation Task	Operation Task Description
Network Planning		network planning goals and the expected network capabilities to be met.
	Data acquisition	Collect related data (such as traffic, performance data, user data, resource utilization) according to data collection rules.
	Data preprocessing	The necessary processing of the collected data before analysis and prediction, such as data review, filter, and classification.
	Network analysis and requirements prediction	According to network planning policies generation results (network construction strategy and network construction standards), and existing performance data and user data, analyze and predict the use of resources on the existing network (such as traffic, performance, user scale, resource utilization), and perform prioritizing, and output a list of planning requirements (such as coverage area, capacity requirements).
	Network planning scheme generation and simulation	Based on the network planning requirements and insight analysis results, output the High Level Planning schemes (network architecture, capacity requirements, and networking solution) and further output the Low Level Planning schemes (such as naming conventions, address planning, number of wireless sites and cells, frequencies, bandwidth, and TAC), and network cutover plan (such as cutover service type, scope, involved network elements, etc.)
	Scheme simulation	Evaluate and simulate the correctness and rationality of the HLP schemes based on the network planning result, and further evaluate the correctness and rationality of the LLP schemes and perform simulation verification.
	Network planning evaluation and decision-making	Comprehensively evaluate HLP schemes and LLP schemes and determine the final HLP scheme and LLP scheme.
	Output network planning scheme	Output the decided high level plan and low level plan for deployment.

5.2.2.3. Deployment flow decomposition and task identification

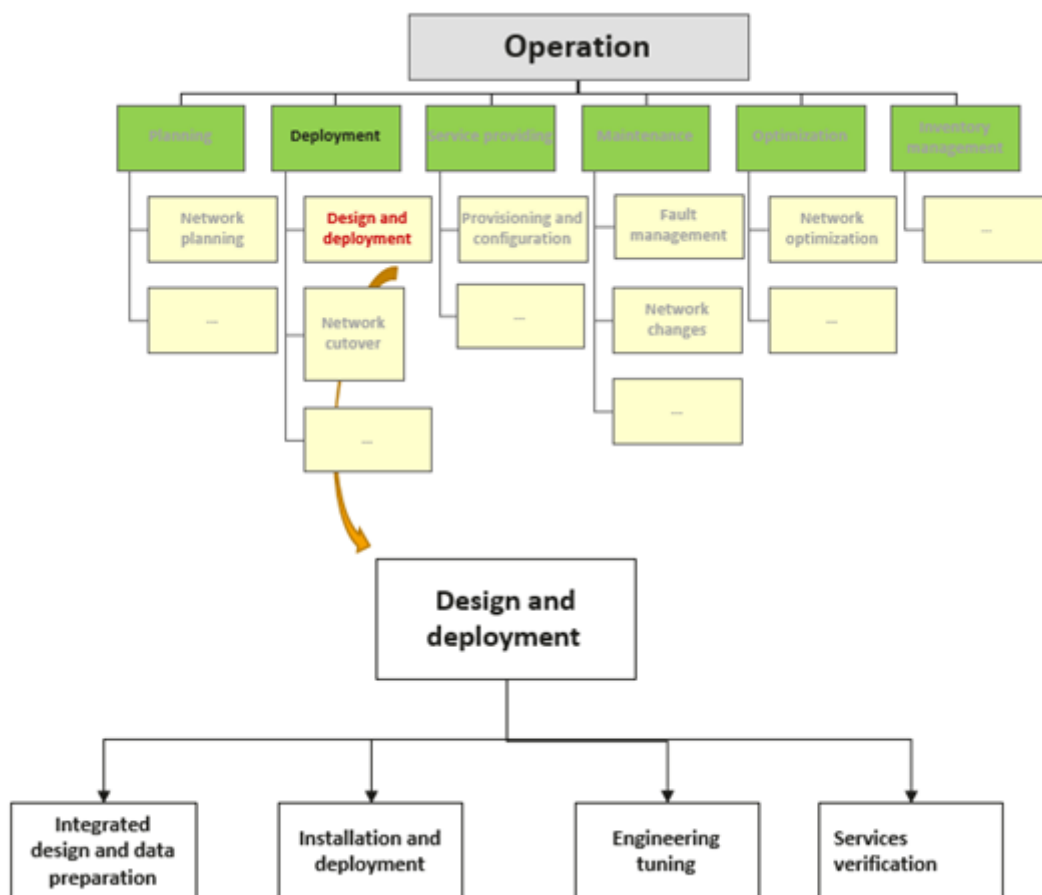


Figure 5-6. Operation tasks of design and deployment

Table 5-4. Operation tasks of design and deployment

Sub Operation Flow	Operation Task	Operation Task Description
Design and Deployment	Integrated design and data preparation	According to the LLD in the planning stage, output the integrated network delivery plan, and prepare network element initialization data, network element parameter data (interface, IP, protocol, etc.) and assurance policy data (link bandwidth/delay/jitter, QoS, Monitoring rules, etc.) and other detailed and full parameters.
	Installation and deployment	Complete the hardware and software installation (traditional network element software installation, VNF network element initialization) according to the network planning output and perform physical acceptance.

Sub Operation Flow	Operation Task	Operation Task Description
	Engineering tuning	Convert the network element data (network element configuration parameters, guarantee policy data) into detailed instructions (such as MML, CLI, API, NETCONF, etc.) and configure it to the network. Execute software and hardware commissioning to ensure that the software and hardware configuration is correct, alarms cleared, or the performance is tuned to be optimal.
	Services verification	Perform services test verification, generate a verification report, pass the acceptance criteria, and meet the specified requirements.

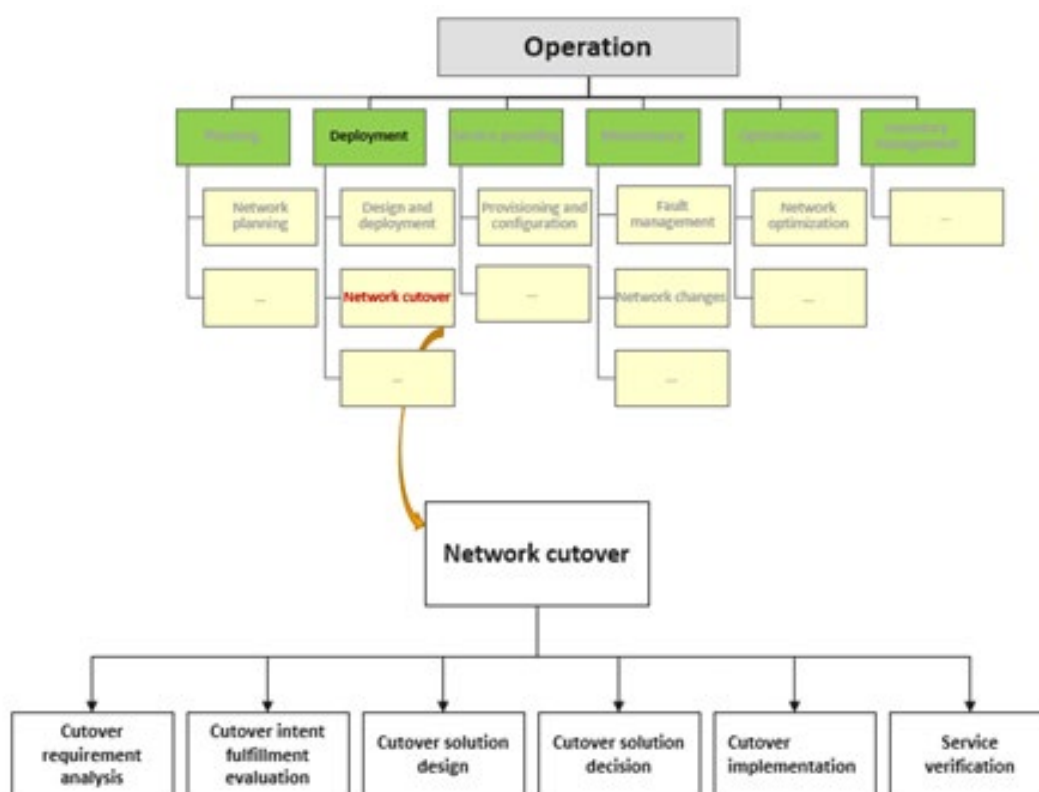


Figure 5-7. Operation tasks of network cutover

Table 5-5. Operation tasks of network cutover

Sub Operation Flow	Operation Task	Operation Task Description
Network Cutover	Cutover requirement analysis	According to the network cutover plan (cutover service type, scope, involved network elements, etc.), evaluate the range of users affected by the cutover, combine the current network service SLA, allowable cutover time period and other factors, and output cutover constraints (such as cutover time window, business interruption time, etc.).
	Cutover intent fulfillment evaluation	Evaluate network cutover intent fulfillment information.
	Cutover solution design(s)	According to the network cutover plan and cutover constraints, complete the cutover solution designs.
	Cutover solution decision	Comprehensively evaluate alternative cutover solutions (such as whether cutover constraints are met, whether other existing network services are affected, etc.) and determine the best solution.
	Cutover implementation	According to the best cutover solution, cutover the network (such as modifying the configuration, path adjustment, UNI port adjustment, etc.).
	Service verification	After the cutover is implemented, verify and confirm the execution results (such as whether the business SLA meets the expected goals, etc.), and perform cutover monitoring, and deal with problems in time.

5.2.2.4. Maintenance flow decomposition and task identification

5.2.2.4.1. Fault management

Fault management is one of the sub operations flows of maintenance. Fault management is decomposed into the following operation tasks. (Note: these tasks mainly cover resource layer and services layer of AN). Faults include multiple types of managed objects, e.g., network alarm, incident of service outages or degradations. Distinguishable from alarm, an incident is identified based on multidimensional data and has service impacts.

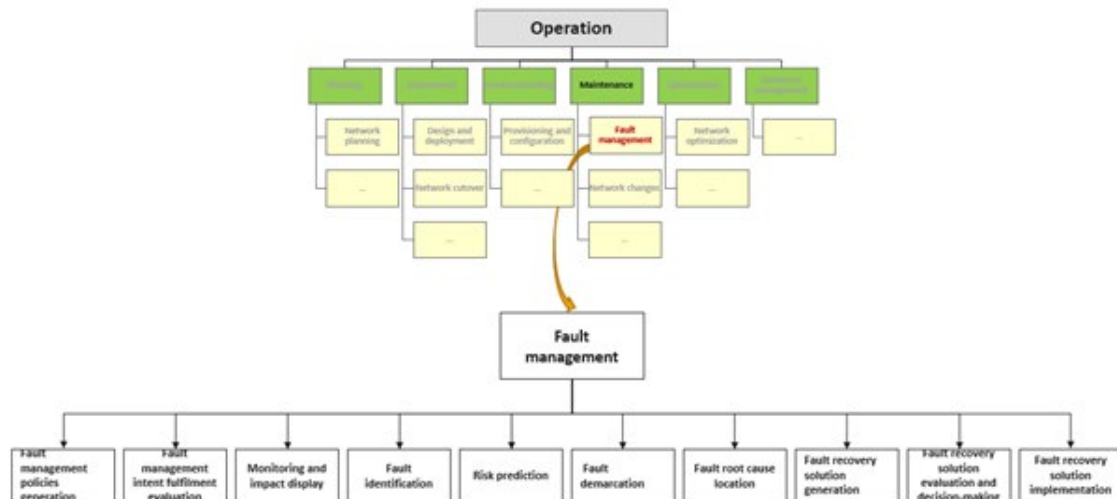


Figure 5-. Operation tasks of fault management

Table 5-6. Operation tasks of fault management

Sub Operation Flow	Operation Task	Operation Task Description
Fault Management	Fault management policies generation	Based on the fault management intent (e.g., routine monitoring requirements, fault recovery requirements etc.) , to generate fault management intent related rules/policies (e.g., monitoring objects, alarm types/levels/filtering rules, alarm reporting strategy, fault recovery policy, etc.) (TBA)
	Fault management intent fulfillment evaluation	The group of tasks of evaluating fault management intent fulfillment information. After fault rectification and risk elimination are performed, verify, and confirm the execution results, such as whether service interruption, quality deterioration, and alarms, KPI exceptions, and incidents are cleared.
	Fault identification	Based on data collection, analyzes network/services operation data and external spatio-temporal data to detect unexpected service interruptions or service quality deterioration in a timely manner.
	Potential fault prediction	Based on data collection, monitors, and analyzes network/services running data and external spatio-temporal data, predicts the development trend of network software and hardware status, and identifies risks that may cause fault in advance, and report it as incident
	Fault impact analysis and display	Based on fault identification and potential fault prediction, analyze and display impacts to service/customer (such as affected base stations and home broadband users).

Sub Operation Flow	Operation Task	Operation Task Description
	Fault Demarcation	Demarcate faults based on the identified faults and potential risks. In the cross-domain scenario, demarcate the fault to a specific technical domain. In the single-domain scenario, demarcate the fault to a specific NE.
	Fault root cause location	Based on the fault demarcation result further locate the specific software and hardware causes (such as configurations, boards, and optical modules) of the fault to support the generation of rectification solutions and rectify services as soon as possible.
	Fault recovery solution generation	Based on the fault demarcation and locating results, generate several alternative recovery/recovery solutions, such as modifying configurations, restarting NEs, replacing boards, and isolating NEs.
	Fault recovery solution selection	Comprehensively evaluate the alternative repair solution (such as whether the solution can solve the fault, whether the repair cost is acceptable, and whether the extra impact on the network) and provide the optimal solution.
	Fault recovery solution implementation	Rectify faults and eliminate potential risks based on the optimal solution. For example, deliver configurations to the network for software faults, isolate NEs or links for hardware faults, or replace or remove boards or optical modules onsite.

5.2.2.4.2. Network change management

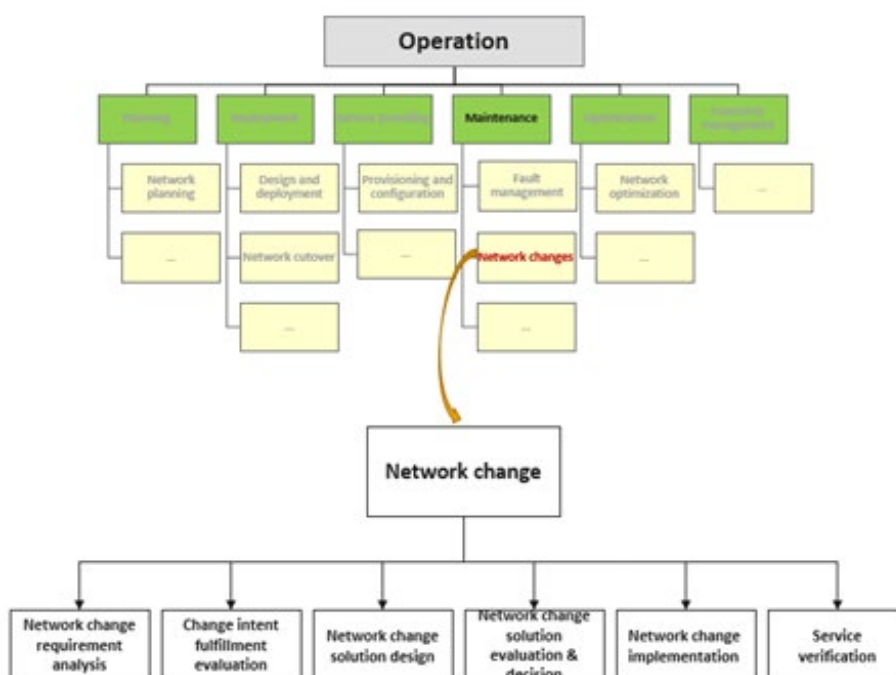


Figure 5-8. Operation tasks of network change

Table 5-7. Operation tasks of network change

Sub Operation Flow	Operation Task	Operation Task Description
Network Change	Network change requirement analysis	According to the network maintenance change request (change type, area range, network elements involved, etc.), evaluate the range of users affected by the change, combine user business SLA, allowable change time period and other factors, and output change constraint (such as change time window, service interruption time, etc.).
	Network change intent fulfillment evaluation	Evaluate the network maintenance change intent fulfillment information.
	Network change solution design	Design of alternative the network maintenance change plans according to the intention and the change constraint output by the services impact assessment.
	Network change solution evaluation & decision	Comprehensively evaluate the alternative network maintenance change plans (such as whether change constraints are met, whether the cost is acceptable, etc.) and give the best plan.

Sub Operation Flow	Operation Task	Operation Task Description
		Evaluate the Maturity at data governance level, in particular for security aspects Role Based Access control, No trust approach within an operation flow. Evaluate safeguards to update security to control configuration changes made within network (from patches to self security with AI).
	Network change implementation	Convert the optimal plan into implementation instructions and implement them to the network and roll back in time when problems occur.
	Service verification	After the network maintenance change is implemented, verify and confirm the implementation results, such as whether the services SLA meets the expected goals

5.2.2.5. Service providing flow decomposition and task identification

5.2.2.5.1. Provisioning and configuration

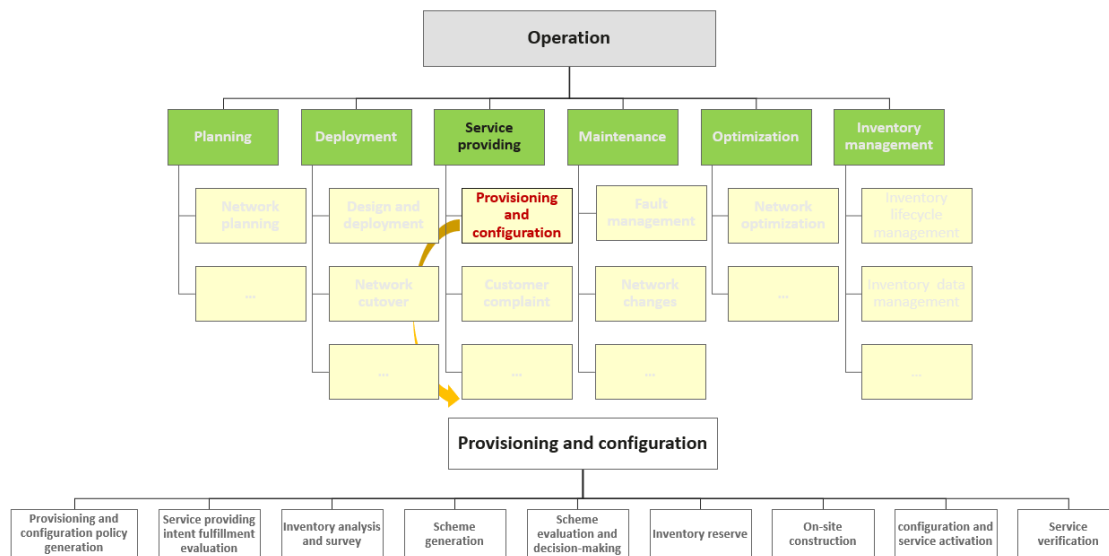


Figure 5-10. Operation tasks of provisioning and configuration

Table 5-8. Operation tasks of provisioning and configuration

Sub Operation Flow	Operation Task	Operation Task Description
	Provisioning and configuration policy generation	Convert user service requirements (such as the number of sites, site location, service volume, and security requirements) to specific network requirements (such as bearer technologies, protection requirements, and NEs (including VNFs lists, security policies, SLA assurance policies, etc.).

Sub Operation Flow	Operation Task	Operation Task Description
Provisioning and Configuration	Service providing intent fulfillment evaluation	Evaluate the satisfaction of the intention.
	Inventory analysis and survey	According to the collected network resources and service data, conduct survey and analysis.
	Scheme generation	Based on the converted network requirements and analysis results, generate provisioning schemes (such as coverage area, deployment location, network topology, and resource allocation).
	Scheme evaluation and decision-making	Comprehensively evaluate the schemes (e.g., whether user requirements are met, whether existing services are affected, and whether resources are sufficient) and provide the evaluation results, output the determined scheme which is to be implemented.
	Inventory reserve	Based on the determined scheme and service requirements, through the workflow engine, the work orders are sent to various relevant departments, and relevant resource(such as network infrastructure resource, transport resource, and pipeline resource) are requested to be reserved by inventory management system , and the relevant process is scheduled and orchestrated.
	On-site construction	Based on the provisioning scheme, carry out on-site construction to ensure that the network is ready on physical resources to support new service and functional requirements, including internal construction, external construction, etc., and carry out quality inspection and acceptance
	configuration and service activation	Convert the decided solution into implementation instructions and deliver the instructions to the network infrastructure, and smoothly activate services. If a problem occurs, roll back the solution in a timely manner.
	Service verification	After the configuration and service activation are implemented, verify and confirm the implementation result (such as service SLA assurance, service connectivity).

5.2.2.5.2. Customer complaint

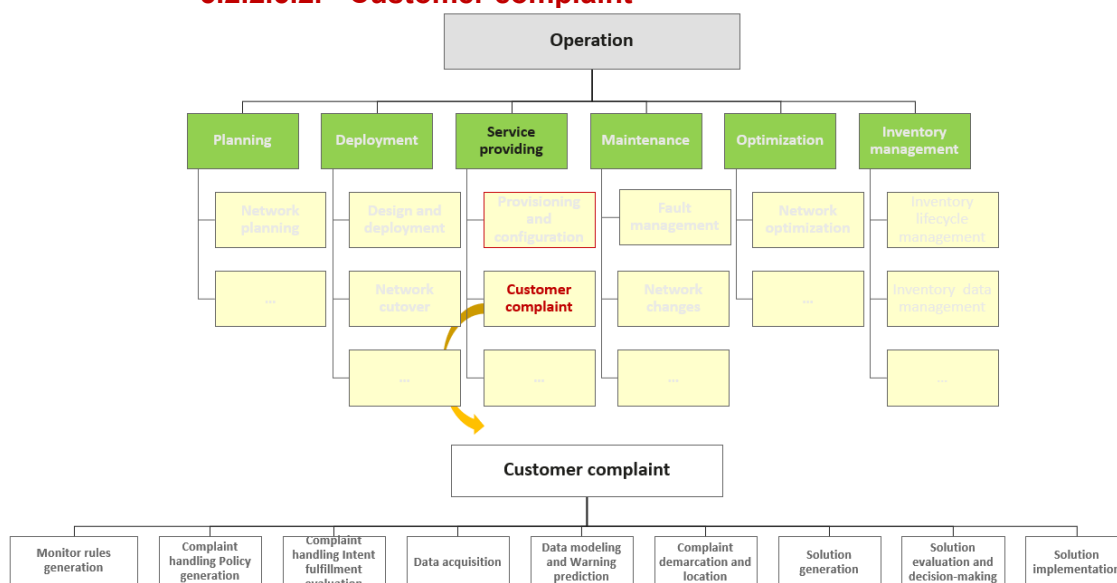


Figure 5-11. Operation tasks of customer complaint

Table 5-9. Operation tasks of customer complaint

Sub Operation Flow	Operation Task	Operation Task Description
Customer Complaint	Monitor rules generation	Generate network monitoring rules based on customer complaint intentions, such as monitoring area, monitoring object, performance data category, data collection cycle, etc.
	Complaint handling Policy generation	Based on customer complaint intentions, generate complaint handling policies, which may include related policies for awareness, analysis, decision, and execution.
	Complaint handling Intent fulfillment evaluation	After the complaint handling solution is implemented, verify the results and evaluate the satisfaction of the intention.
	Data acquisition	Collect kinds of service data based on defined data collection rules.
	Data modeling and Warning prediction	Based on the collected service data, an analysis model required to predict customer complaints can be generated, and customer complaints can be predicted.
	Complaint demarcation and location	Based on the acquired monitoring, performance, service quality and other data analysis to realize the demarcation and location of complaints.
	Solution generation	Based on the analysis results of the demarcation and location of the complaints, generate several alternative solutions .

Sub Operation Flow	Operation Task	Operation Task Description
	Solution evaluation and decision-making	Evaluate the feasibility of each complaint handling solutions and predict the recovery effect and direct/indirect impact, etc., and determine the optimal solution based on the evaluation result.
	Solution implementation	Implementation of customer complaint solution.

Note : The content related to (awareness, analysis, decision, execution, intent/experience) is subject to change to keep consistent with IG1218, IG1253 in the future version.

5.2.2.6. Inventory management flow decomposition and task identification

5.2.2.6.1. inventory lifecycle management

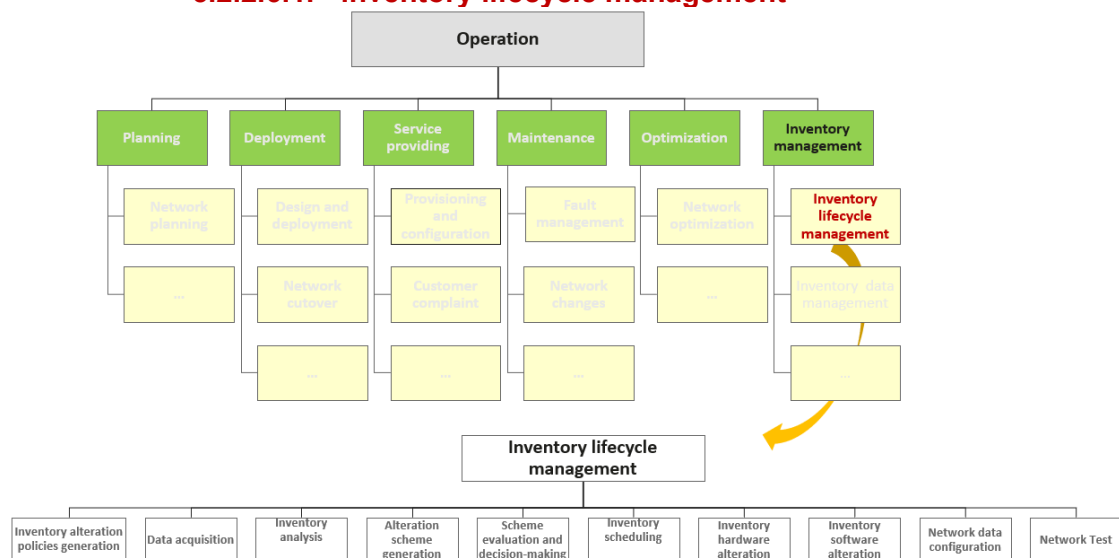
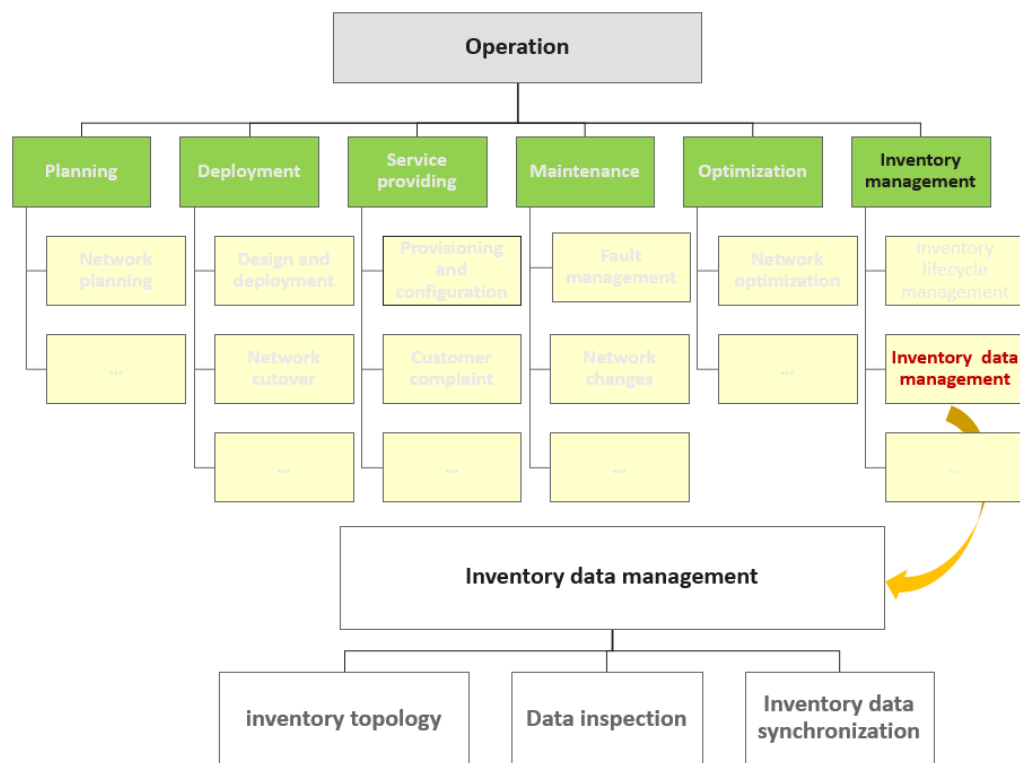


Figure 5-12. Operation tasks of inventory lifecycle management

Table 5-10. Operation tasks of inventory lifecycle management

Sub Operation Flow	Operation Task	Task Description
	Inventory alteration policies generation	Based on the inventory alteration intention(such as resource status and alteration requirements), output the alteration strategy.
	Data acquisition	Collect resource data based on defined data collection rules.
	Inventory analysis	Based on the collected resource data, check the network status before the change and output the survey and analysis.

Sub Operation Flow	Operation Task	Task Description
Inventory Lifecycle Management	Alteration scheme generation	Based on alteration policies and generated analysis results, design alteration schemes.
	Scheme evaluation and decision-making	Comprehensively evaluate the schemes (e.g., whether user requirements are met, whether existing services are affected, and whether resources are sufficient) and provide the evaluation results, output the determined scheme which is to be implemented.
	Inventory scheduling	Based on the determined scheme and alteration requirement, inventory are scheduled.
	Inventory hardware alteration	According to the inventory scheduling result, reserve hardware resources, and cooperate with other OSS system to make changes to the hardware facilities of the system, monitor and report on the usage and access to the hardware resource in inventory .
	Inventory software alteration	According to the inventory scheduling result, reserve software resources, and cooperate with other OSS system to upgrade software, monitor, and report on the usage and access to the software resource in inventory .
	Network data configuration	According to the inventory scheduling result, cooperate with other OSS system to configure network, monitor and report on the usage and access to the physical and logical resources in inventory .
	Network test	After the implementation of alteration and configuration, cooperate with other OSS systems(such as orchestration system, network dashboards) to test the network KPI to ensure the normal operation of the network in inventory.

5.2.2.6.2. inventory data management**Figure 5-13. Operation tasks of inventory data management****Table 5-11. Operation tasks of inventory data management**

Sub Operation flow	Operation Task	Task Description
Inventory Data Management	Inventory topology	The internal topological relationships of data resources can be queried and can realize automatic maintenance and analysis in higher levels.
	Data inspection	Analyze, review, and inspect resource data
	Inventory data synchronization	Routine operations of resource data, add, delete, modify, and check

Note : The content related to (awareness, analysis, decision, execution, intent /experience) is subject to change to keep consistent with IG1218, IG1253 in the future version.

5.2.2.7. Network optimization flow decomposition and task identification

The optimization flow is decomposed with the following operation tasks. (Note: these tasks mainly cover resource layer and services layer of AN)

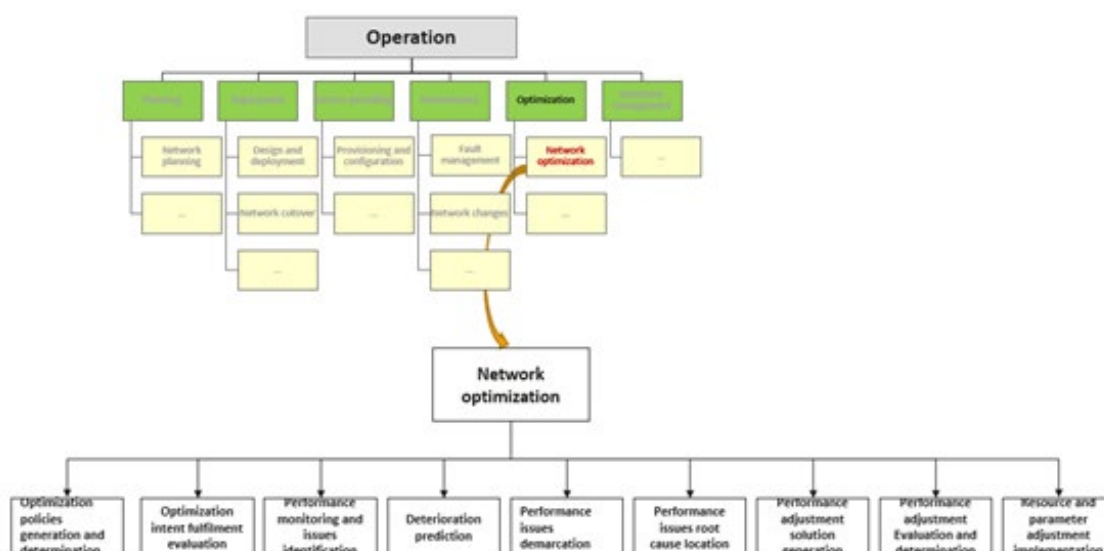


Figure 5.14 Operation tasks of network Optimization

Table 5-12: Operation tasks of network Optimization

Sub Operation Flow	Operation Task	Operation Task Description
Network Optimization	Optimization policies generation and determination	Based on the optimization intent (e.g., resource utilization, user experience, energy saving requirements, public opinion analysis, and proactive user planning/improvement requirements), determine optimization intent related rules/policies (e.g., monitoring rules, issue identification policies, issue analysis policies and parameters adjustment policies). (TBA)
	Optimization intent fulfillment evaluation	Optimization intent fulfillment evaluation, after the optimization action is executed, verify and confirm the execution effect, such as whether the customer experience, energy saving requirements, and resource utilization meet the requirements.
	Performance monitoring and issues identification	Based on information collection, monitors and analyzes network/services running data and external spatiotemporal data to identify issues to be optimized that affect customer experience (such as weak coverage areas), improper resource usage (excessive energy consumption and unbalanced resource load), and proactive performance issues identification.
	Deterioration issue prediction	Analyzes network/services running data and external spatiotemporal data, predicts performance and resource utilization trends, and identifies risks (such as insufficient capacity and insufficient licenses) that may affect customer experience in advance, and report it as incident.

Sub Operation Flow	Operation Task	Operation Task Description
	Performance issue demarcation	Demarcate performance issues based on the identified performance abnormality or deterioration prediction information and environment monitoring. In cross-domain scenarios, demarcate the issues to specific technical domains (such as wireless, transport, and core networks). In single-domain scenarios, demarcate the -issues to specific optimization objects (such as NEs).
	Performance issue root cause location	Based on the performance issues demarcation result, locate the software and hardware causes (such as configuration, air conditioner, and power environment) that cause performance abnormality or deterioration to support optimization solution generation.
	Performance adjustment solution generation	Generate several alternative parameter adjustment solutions (such as software parameter modification and hardware adjustment) based on the performance issues demarcation and root cause locating results
	Performance adjustment solution selection	Comprehensive evaluation of alternative adjustment solutions (e.g., whether customer experience is affected, whether the adjustment solution meets the optimization objective, and whether the adjustment cost is acceptable) and give the optimal scheme.
	Resource and parameter adjustment implementation	Performs optimization based on the optimal solution and delivers the optimized resource and parameter configurations to the network through CLI, MML, NETCONF, or other interfaces.

5.2.3. Task contextualization /minimal evaluation unit identification

The standard does not specify specific task contextualization. Users are supposed to refine task contextualization according to their own major networks or services as the smallest unit for evaluation.

When selecting a task contextualization, the developing network elements and services shall be given priority, to ensure consistency with the L0-L5 technology maturity evolution. The task contextualization shall lead to automation and intelligence. It is not necessary to cover all situations.

5.3. Level Evaluation

5.3.1. General technical capability level evaluation

Qualitative assessment

A clear, unambiguous and operable detailed definition of general technical capability requirements for all levels should be given to all the tasks according to the technical maturity model, with regards to the selected context. The technical maturity level is determined via the qualitative assessment.

As shown in the table, taking the task “Fault identification” and “Demarcation and locating” as examples. From the network dimension, IP network service performance deteriorates can be selected as task contextualization example in the IP domain.

Table 5-13. Example for qualitative assessment

Task	Task Contextualization Example	L0	L1	L2	L3	L4	L5
Fault Identification	IP network service performance deteriorates	Full manual off-line operation	Work order recording. Tool/system data collection. Manual fault identification;	On the alarm or event monitoring page, the following two types of information are displayed: 1. Alarms or events derived from the original alarms about link	Check the derived alarm or event configuration page, and set the following parameters: 1. Alarm title or event title. 2. Derivative conditions or event	By artificial training of predictive intelligent AI model, instead of manually configured rules, the system associates multidimensional data through predictive intelligence, and intelligently	Fully automatic intelligent identification of fault events or deterioration events without human involvement

Task	Task Contextualization Example	L0	L1	L2	L3	L4	L5
				performance deterioration. 2. Information about the alarm link or port.	identification rules. 3. Associated alarm scope.	identifies fault events or deterioration events. When the AI model needs to be iterated, it can be trained and loaded manually.	
Demarcation and Locating	IP network service performance deteriorates	Full manual off-line operation	Work order recording. Human use of tools/systems assisted demarcation (e.g., connectivity testing, etc.). Manual impact analysis;	Check event scenario page and the following information is displayed: 1. alarms related to the fault device. 2. Engineering information of the faulty device.	On the O&M capability configuration page, you can configure the following: 1. Check the alarm capability of the device. 2. Verify the engineering information capability.	Instead of manually configured rules, artificial AI models are trained, and the system uses diagnostic intelligence to determine the fault location or intelligence to accurately analyze the service impact. When the AI model needs to be iterated, it can be trained	Fully automatic fault location or intelligent service impact analysis without human involvement

Task	Task Contextualization Example	L0	L1	L2	L3	L4	L5
						and loaded manually.	

5.3.1.2. Quantitative Aggregation

The quantitative aggregation is then performed after the qualitative assessment. The score of each task is obtained by aggregating and averaging the qualitative assessment scores of all the contextualization involved in.

The quantitative aggregation method includes the following steps:

- qualitatively evaluate the task contextualization and get the score
- calculate the average level of the included task contextualization as the level of the task ;
- calculate the average level of the included tasks as the level of the operation flow.

Take the IP-fault identification as an example.

Table 5-14. Example for quantitative aggregation

Service/network	Task	Task Contextualization	Task Evaluation Criteria	Score
IP	Fault identification	1	L1 : ... L2: ... L3:	2
		2	L1 : ... L2: ... L3:	3

Service/network	Task	Task Contextualization	Task Evaluation Criteria	Score
				(2+3)/2=2.5

5.3.2. Resource closed loop level evaluation

5.3.2.1. Interface requirements

According to section 4.2.3.2, resource closed loops should be aligned to the interface requirements of general technical capabilities in order to support upper layer's planning, deployment, maintenance, optimization, and service providing activities. The general requirements as specified in Table 4-3 need to be adapted to the context of specific tasks. Below is an example of interfaces requirements to resource closed loop derived from table 5-13 in Section 5.3.1.1.

Table 5-15. Example of interfaces requirements

Task	Task Contextualization Example	Interface Requirements -L0	Interface Requirements-L1	Interface Requirements -L2	Interface Requirements-L3	Interface Requirements-L4	Interface Requirements-L5
Fault Identification	IP network service performance deteriorates	Provide manual configuration interface	Provide interfaces for configuring data collection types and modes	Provide the interface of alarm/event management derived from the original alarms of link performance deterioration	Provide flexible and customizable interface for derived alarm/event management configuration.	Provide necessary AI model management interfaces, such as training data acquisition, model deployment, model training, etc. in the scenario of centralized and distributed AI model application.	Provide self-evolving capability interfaces in response to closed-loop management requirements such as automatic generation, application and update of knowledge and AI/ML models

Task	Task Contextualization Example	Interface Requirements -L0	Interface Requirements-L1	Interface Requirements -L2	Interface Requirements-L3	Interface Requirements-L4	Interface Requirements-L5
Demarcation and Locating	IP network service performance deteriorates	Provide manual configuration interface	Provide Interfaces to configure assistant capability of demarcation and location	The interface provides alarm verification information and engineering information about the fault device.	Provides flexible and customizable interface for derived alarm/event management configuration.	Provide necessary AI model management interfaces, such as training data acquisition, model deployment, model training, etc in the scenario of centralized and distributed AI model application.	Provide self-evolving capability interfaces in response to closed-loop management requirements such as automatic generation, application and update of knowledge and AI/ML models

5.3.2.2. Scoring methods

5.3.2.2.1. Tasks categorization in the context of resource closed loop

In the context of resource closed loop, control-loops are used to describe (and execute) the necessary steps to fulfill certain management purposes. An operation flow is an encapsulation of one or more management tasks that exist in the context of a control loop. The autonomy capabilities of the tasks within the operation flow may impact the network autonomy level. The following are the potential categorization of the tasks in an operation flow:

- **Intent/Experience:** The group of tasks which translate intent from operator or customer into detailed operations which may affect one or more of the following groups of tasks (i.e., awareness, analysis, decision, execution), also evaluate and feedback intent fulfillment information (e.g., the intent is satisfied or not).
- **Awareness:** The group of tasks that monitor and understand what is happening or exists in quality/state of network, such as collecting network information, preprocess information and selective notification, etc.
- **Analysis:** The group of tasks which analyze data generated in the awareness phase, use technologies (such as model inference) and analysis, make a recommendation for decision.

- **Decision:** The group of tasks that decide the necessary management operation for execution, e.g., network configuration or adjustment.
- **Execution:** The group of tasks which execute the management operations and feedback the result to the intent requester.

For example, the relationship between tasks and AI-based cognitive workflow in monitoring and troubleshooting operation process is as follows:

- A task group of fault management policies generation and fault management intent fulfillment evaluation constitutes intent.
- A task group of fault impact analysis, display, fault identification, and risk prediction constitutes awareness.
- A task group of fault demarcation fault root cause location and fault recovery solution generation constitutes analysis.
- A task of fault recovery solution selection constitutes decision.
- A task group of solution implementation and verification constitutes execution.

The principles for decomposing operation flow to operation task are as follows:

- **Completeness:** All the O&M must be included in the operation tasks.
- **Balance:** The sizing and granularity of each task is relatively balanced.
- **Non-Overlapping:** The operation task is atomic and does not have functional overlaps with other tasks.

5.3.2.2.2. Selected scenario in the context of resource closed loop

Selected scenario is the subdivided work scope of an evaluation object in the context of resource closed loop. When the evaluation object is at L1 to L4, the intelligent capability only needs to cover selected scenarios. In Figure 5-1, a 3D model is established according to the three dimensions of operation flow domain, network domain and service domain to determine the evaluation object. In practice, resource closed loop evaluation process needs to be carried out in a specific scenario, which means the typical scenarios should be further refined according to the network domain.

The example in figure 5-15 depicts the relationship between operation flow, sub-operation flow, network domain, and selected scenarios. The selected scenarios are identified by the intersection of sub operation flow with the network technologies (OTN and IP as examples). For the intersection ONT – Fault management, selected scenarios can be NE unreachable, or abnormal link, etc.

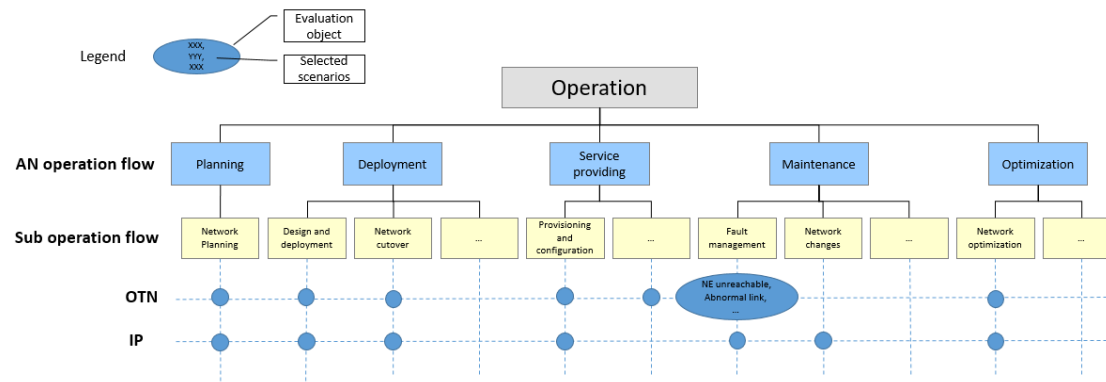


Figure 5-15 An example of selected scenarios

Principles of selected scenarios when evaluating resource closed loops

When evaluating resource closed loop, selected scenarios follow below 2 principles:

1. **Independence:** For a given evaluation object, scenarios are independent and decoupled. There is no overlap or intersection between scenarios.
2. **Purpose oriented:** A selected scenario should include all necessary task(s) to achieve certain management and control purpose.

As described in section 5.2, resource closed loop supports upper layer planning, deployment, maintenance, optimization, service providing and inventory management activities. Scenarios can be categorized based on these activities, examples shown in figure 5-16:

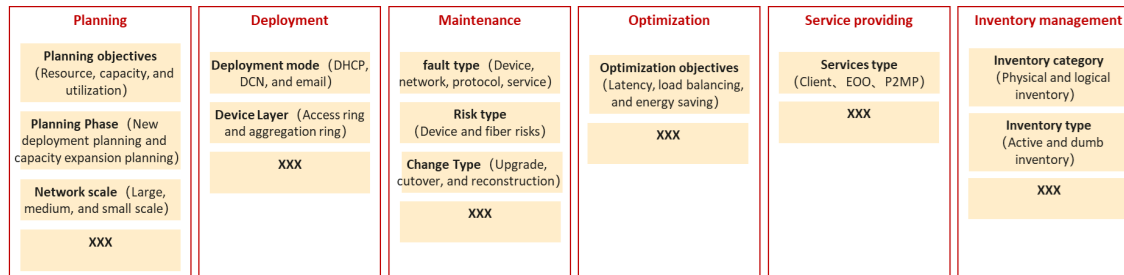


Figure 5-16. Categories of autonomous network level scenarios

- (1) Scenarios for planning operation flow may be classified as planning objects, planning phase and network phase, etc.
- (2) Scenarios for deployment operation flow may be classified as deployment mode and device layer, etc.
- (3) Scenarios for maintenance operation flow may be classified as fault type, risk type and change type, etc.
- (4) Scenarios for optimization operation flow may be classified as optimization objectives, etc.
- (5) Scenarios for service providing operation flow may be classified as Resource Facing Service type, etc.
- (6) Scenarios for inventory management operation flow may be classified as inventory category and type, and so on.
- (7) In addition to above scenario categories, a common dimension combines with network operation flows to generate particular selected scenarios. The common dimension can be network RAT (e.g. 4G, 5G), networking type (e.g. IPRAN, SPN, OTN, etc.), and spatial location (e.g. indoor, outdoor), etc.

Each network domain can define its domain-specific selected scenarios according to these categories.

5.3.2.2.3. Scoring method 1

This method focuses on the autonomy capability of selected scenarios in a resource closed loop and provides leads to improve given scenarios.

1. Qualitatively evaluate all tasks in all selected scenarios of the evaluation object. Referring to table 4-1, evaluating the resource closed loop pays attention to the system's capabilities, a key criterion for determining the intelligence level of each operation task in L0 to L5 is to consider what agents, human, machine, or both, are performing the task. Specific scoring rules can be formulated for different technology domains. The score of a task is the highest level that the task capability meets corresponding requirement.
2. Score of a selected scenario is the weighted average of scores of all tasks in that scenario.

3. AN Level of the evaluation object is the weighted average of scores of all selected scenarios in that evaluation object.

Below is an example of scoring an evaluation object IP Network – fault management.

Tasks involved in IP network -Fault management are

- Intent/experience task
 - A1 Intent interpretation : The task is to understand the monitoring and assurance intention and interpret the intention to the system monitoring rules, policies, or functions.
- Awareness tasks
 - B1 Visualized monitoring: The task is to create E2E SLA monitoring, service traffic monitoring, and status and traffic monitoring of network nodes and links on the service path, after an IP network service is provisioned.
 - B2 Fault identification: The task is to identify network exceptions. Alarms are compressed and aggregated to implement identification of massive faults.
- Analysis tasks
 - C1 Demarcation and locating: The task is to perform hop-by-hop diagnosis to quickly demarcate IP network services, associates service- and network-level problems, and match fault modes, such as the smallest replaceable unit can be located.
 - C2 Solution generation: The task is to calculate alternative paths based on path computation policies and service SLA requirements, after detecting a service fault.
- Decision task
 - D1 Evaluation and decision-making: The task is to evaluate the repair solution, provide the evaluation result, and determine the optimal
- Execution tasks
 - E1 Solution implementation: The task is to perform service path adjustment.
 - E2 Service verification: The task is to perform service acceptance and check whether the damaged service is restored, including whether the service connectivity and SLA meet requirements.

The autonomy level score of above tasks can be evaluated according to below table 5-16.

Task category	Task	L0	L1	L2	L3	L4	L5
Execution	E1 Solution implementation	P	P/S	S	S	S	S
	E2 Service verification	P	P/S	P/S	S	S	S
Awareness	B1 Visualized monitoring	P	P/S	P/S	S	S	S
	B2 Fault identification	P	P/S	P/S	S	S	S
Analysis	C1 Demarcation and locating	P	P	P/S	S	S	S
	C2 Solution generation	P	P	P/S	S	S	S
Decision	D1 Evaluation and decision-making	P	P	P	P/S	S	S
Intent/Experience	A1 Intent interpretation	P	P	P	P	P/S	S

P: Person

P/S: Person + System

S: System

Table 5-16 IP network fault management tasks autonomy levels example

Note: Table 5-16 is only an example. Evaluation criteria should refer to specification developed by network domain specific SDO. For a task, requirements for 2 levels marked with same color and letter may vary slightly. If a task meets the requirements of two or more levels, the highest level is the score. Users or other SDOs may have their own rules.

According to typical IP network deployment practice, table 5-17 gives some examples of selected scenario.

Table 5-17 Example of selected scenarios for IP network – Fault management

	Selected Scenario 1	Selected Scenario 2	Selected Scenario 3
Fault Type	NE unreachable	Abnormal link	VPN fault

Based on the score of tasks, score of selected scenarios and the overall AN Level of the evaluation object can be derived (table 5-18). For simplicity, all tasks are assigned identical weights, so are all scenarios.

Table 5-18 ANL calculation of IP network – Fault management example

Task Category	Tasks	Scenario 1 NE unreachable	Scenario 2 Abnormal link	Scenario 3 VPN fault
Intent	A1	3	3	3
Awareness	B1	4	4	2
	B2	4	4	2
Analysis	C1	4	2	2
	C2	3	2	1
Decision	D1	2	3	3
Execution	E1	4	4	4
	E2	4	2	2
Score of Selected Scenarios		3.51	3.01	2.38
AN Level of the Evaluation Object		2.96		

5.3.2.2.4. Scoring method 2

This method focuses on identifying tasks of the lowest autonomy level and urge improvement of weaknesses.

1. For a single selected scenario, the highest level that all involved tasks meet respective requirements, is the basic level score, which is an integer, denoted as L_{base} .

Using the same example evaluation object as in above section, Table 5-19 is created for the selected scenario1 NE unreachable. If a task meets the requirement of a certain level, corresponding cell in the table is marked 1, otherwise marked 0. The scenario meets all requirements of L2, the A1 and D1 don't meet the requirements of L3. Thus, L_{base1} of this scenario is 2

Task category	Task	Weight	L0	L1	L2	L3	L4
Intent/ Experience	A1	10%	1	1	1	1	0
Awareness	B1	20%	1	1	1	1	1
	B2	15%	1	1	1	1	1
Analysis	C1	10%	1	1	1	1	1
	C2	10%	1	1	1	1	0
Decision	D1	15%	1	1	1	0	0
Execution	E1	10%	1	1	1	1	1
	E2	10%	1	1	1	1	1

Table 5-19 Example: basic level score of a selected scenario

2. The extra level score of a selected scenario is determined by tasks that satisfy the requirements for the next higher (i.e., $L_{base} + 1$) level. The extra score is a fraction or decimal, denoted as L_{extra} .

$$L_{extra} = \sum_{i=1}^n W_{ti} L_{ti}$$

Where $0 < W_{ti} \leq 100\%$ and $\sum W_{ti} = 100\%$ is the weight of task i which is decided by the user, n is the number of tasks included in the scenario, L_{ti} is the level score of task i , determined by following equation.

$$L_{ti} = \begin{cases} 1, & \text{if task } i \text{ meets the requirement for the level } (L_{base} + 1) \\ 0, & \text{otherwise} \end{cases}$$

In the above example,

$$L_{extra1} = 10\% * 1 + 20\% * 1 + 15\% * 1 + 10\% * 1 + 10\% * 1 + 15\% * 0 + 10\% * 1 + 10\% * 1 = 0.85$$

3. The level score of a selected scenario is the sum of its basic level score and extra level score.

$$L_{scenario} = L_{base} + L_{extra}$$

$$\text{In the above example, } L_{scenario1} = L_{base1} + L_{extra1} = 2 + 0.85 = 2.85$$

Following the same approach, we can get $L_{scenario2} = 2.7$, and $L_{scenario3} = 1.9$

4. The final score of an evaluation object is the weighted average of the level scores of all involved selected scenarios.

Assuming that an evaluation object consists of m selected scenarios, the final level score is

$$L = \sum_{i=1}^m W_{si} L_{si}$$

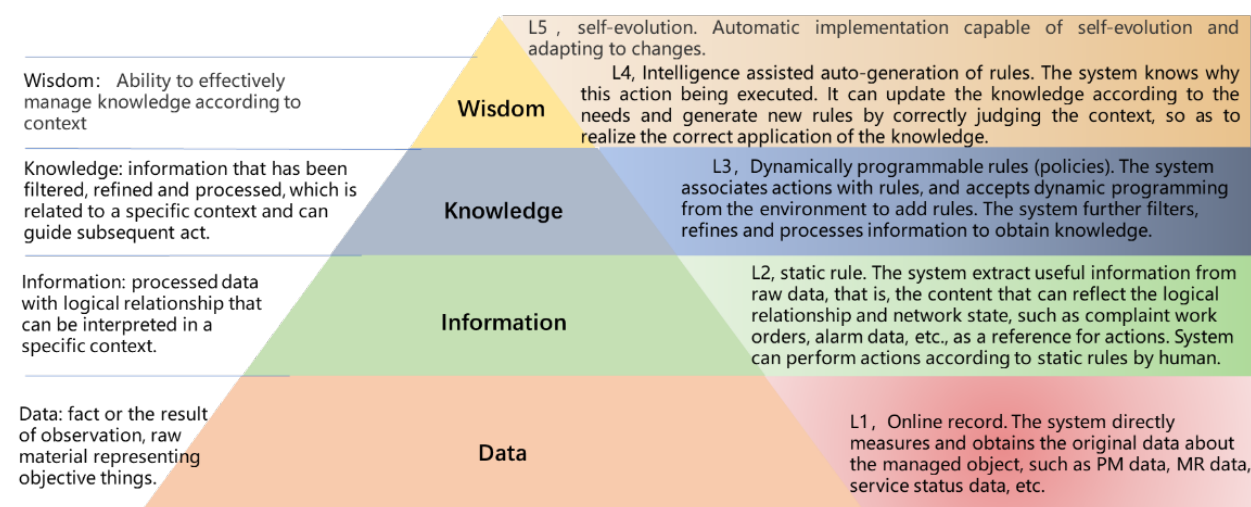
Where, m is the number of selected scenarios included in the evaluation object, $0 < W_{si} \leq 100\%$ and $\sum W_s = 100\%$ is the weight of scenario i , L_{si} is the level score of scenario i .

In the example, Let $W_{s1} = 40\%$, $W_{s2} = 40\%$, and $W_{s3} = 20\%$, the final level is $L = 40\% * 2.85 + 40\% * 2.7 + 20\% * 1.9 = 2.6$

6. References

#	Title	Organization
1	Taxonomy and Definitions for Terms Related to Driving Automation Systems for On-Road Motor Vehicles SAE J3016, April 2021	SAE
2	A model for types and levels of human interaction with automation IEEE Transactions on Systems Man and Cybernetics - Part A Systems and Humans Parasuraman, Sheridan, & Wickens	IEEE Paper
3	IG1230 Autonomous Networks Technical Architecture V1.0.0	TM Forum
4	Management support systems: towards integrated knowledge management	Human systems management
5	From data to wisdom	Journal of applied systems analysis
6	IG1256 Autonomous Network Effectiveness Indicators v1.0.0	TM Forum
7	IG1258 Autonomous Networks Glossary v1.1.0	TM Forum
8	IG1218 Autonomous Networks Business Requirements and Framework v2.2.0	TM Forum

7. Appendix A: DIKW



8. Administrative Appendix

8.1. Document History

8.1.1. Version History

Version Number	Date Modified	Modified by:	Description of changes
0.1 - 0.8	05-May-2021	Wang Xu James O'Sullivan Feng Chengcheng	Initial draft, several iterations
0.9	10-May-2021	Wang Xu James O'Sullivan Feng Chengcheng	Edits after TA meeting held 5 th May Updates to section 6 to reflect changes in section 4 and 5 Various reformatting
0.91	19-May-2021	James O'Sullivan	Applying minor updates
0.92	25-May-2021	Wang Xu	Minor editorial updates
1.0.0	28-May-2021	Alan Pope	Final edits prior to publication. Incorporated all sections on a single Confluence page.
1.1.0	30-Jul-2021	Wang Xu James O'Sullivan Feng Chengcheng	Updates for sprint 4, elaborating deployment and inventory mgt flows adding subsections to maintenance flows diagram updates
1.2.0	16-Jun-2023	Feng Chengcheng Hui Li	Updates to text in all Sections. Added 'Resource closed loop evaluation method' example and 'Scoring method' and associated formulae.

8.1.2. Release History

Release Status	Date Modified	Modified by:	Description of changes
Pre-production	28-May-2021	Alan Pope	Initial release
Production	26-Jul-2021	Adrienne Walcott	Updated to reflect TM Forum Approved status
Pre-production	30-Jul-2021	Alan Pope	Updated to v1.1.0
Production	17-Sep-2021	Adrienne Walcott	Updated to reflect TM Forum Approved Status

Release Status	Date Modified	Modified by:	Description of changes
Pre-production	16-Jun-2023	Alan Pope	Updated to v1.2.0
Production	04-Aug-2023	Adrienne Walcott	Updated to reflect TM Forum Approved status

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